AMERICAN ACADEMY OF OPHTHALMOLOGY®

Long-Shaft Vitrectomy Probe for Vitreoretinal Diseases in Highly Myopic Eyes: A Randomized Controlled Trial

Yun Hsia, MD, MS,[1] *Cheng-Yung Lee, MD,*[1,2] *Mei-Chi Tsui, MD,*[1,3] *Shih-Wen Wang, MD,*[1,4] *Chien-Jung Huang, MD,*[1,5] *I-Hsin Ma, MD, MS,*[1,2] *Kuo-Chi Hung, MD,*[1,6] *Muh-Shy Chen, MD, PhD,*[1,7] *Zih-Wei Yang, MD,*[1] *Bo-Da Huang, MD,*[1] *Ting-Chieh Ko, MD,*[1] *Tzyy-Chang Ho, MD*[1]

Objective: To evaluate the safety and efficacy of a long-shaft vitrectomy probe in highly myopic eyes undergoing pars plana vitrectomy.

Design: A randomized controlled trial.

Subjects: Highly myopic eyes (axial length [AL] >26 mm) with epiretinal membrane (ERM), myopic tractional maculopathy, and retinal detachment.

Methods: The enrolled eyes were randomly assigned to a study group (UltraVit 25 Ga, 31.75 mm, Alcon) and a control group (UltraVit 25 Ga+, 27 mm, Alcon). Stratified randomization was performed to balance the proportion of eyes with AL >28 mm between groups.

Main Outcome Measures: Trocar removal rate and core vitrectomy time were assessed as primary outcomes, and instrument bending as a secondary outcome. Anatomical and visual outcomes and complications were documented for 6 months. Subgroup analysis was performed to compare the eyes with AL >28 mm to those without.

Results: We included 86 patients with a mean age of 60.7 ± 9.6 years and an AL of 29.15 ± 2.14 mm. Two groups had comparable core vitrectomy times (−0.5 minutes, $P = 0.172$). The study group had a lower trocar removal rate (5% vs. 67%, $P < 0.001$) but a higher instrument bending rate (36% vs. 14%, $P = 0.036$), particularly in eyes with AL >28 mm. In eyes with AL >28 mm, the standard vitrectomy probe faced a significantly greater difficulty in cortical vitreous removal, internal limiting membrane (ILM) trimming, or posterior vitreous detachment induction compared with the long-shaft vitrectomy ($P < 0.001$). At 6 months, significant visual improvement (logarithm of the minimum angle of resolution) and anatomical success were achieved (study: −0.22, 95%; control: −0.24, 88%). Eyes with ERM in the study group, not the controls, had significant visual improvement (−0.21, $P = 0.02$ vs. −0.09, $P = 0.34$).

Conclusions: The long-shaft vitrectomy probe is safe and efficient, especially in highly myopic eyes with AL >28 mm, despite a higher instrument bending rate. It provides improved access to the posterior pole, allowing for delicate removal of adherent cortical vitreous and trimming of ILM flaps. Addressing increased instrument bending due to the sleeveless design is important for future design.

Financial Disclosure(s): The author(s) have no proprietary or commercial interest in any materials discussed in this article. *Ophthalmology Science 2025;5:100824* *© 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Supplemental material available at www.ophthalmologyscience.org.

The prevalence of high myopia is on the rise,[1,2] and these eyes are at risk of pathologic myopia and vision-threatening complications, including cataract, glaucoma, and macular degeneration.[3–5] Some of the complications require surgical intervention, such as macular hole (MH), epiretinal membrane (ERM), retinal detachment (RD), and myopic tractional maculopathy (MTM).[3,5–8]

Pars plana vitrectomy (PPV) treats vitreoretinal diseases by removing the vitreous traction and using diverse techniques of internal limiting membrane (ILM) peeling and flap.[9] However, performing PPV in highly myopic eyes is challenging due to elongation of the eyeballs. Standard instruments may be too short to complete premacular membrane and posterior hyaloid removal, trimming of the ILM flap, and fluid–air exchange.[9,10] Surgeons need to verticalize instruments to reach the posterior pole, which causes scleral indentation and bending of the instrument. Trocar removal may be necessary to gain extra effective length of the instruments.

Altered ultrastructures of the sclera,[11,12] intraoperative stretching of the sclera wound, and removal of the trocar increase the risk of sclerotomy leakage and early postoperative hypotony.[9] Modified surgical instruments, such as long-shaft forceps, cannulas, and curved elongated instruments, were proposed.[13,14] The accessibility of the posterior pole for the vitrectomy probe is crucial during

© 2025 by the American Academy of Ophthalmology
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/). Published by Elsevier Inc.

https://doi.org/10.1016/j.xops.2025.100824
ISSN 2666-9145/25

critical steps such as inducing posterior vitreous detachment, removing cortical vitreous, or trimming the ILM flap. However, the commonly used 25-gauge vitrectomy probe (27 mm, UltraVit 25+ Ga, Alcon) is shorter than the 20- and 23-gauge probes (31.75 mm, UltraVit 20 Ga and 23 Ga, Alcon). Therefore, this study aims to investigate the efficacy and safety of an alternative 25-gauge vitrectomy probe with a longer shaft length of 31.75 mm in highly myopic eyes.

Methods

This prospective, double-masked, randomized controlled trial was conducted at the National Taiwan University Hospital between July 2021 and February 2023. The enrolled patients had axial length (AL) >26 mm or refractive error less than −6.0 D and required PPV for vitreoretinal disease. Exclusion criteria included (1) previous vitreoretinal surgery and ocular surgeries associated with conjunctiva scarring, (2) history of ocular penetrating trauma, (3) connective tissue disease, and (4) need for commitment to encircling buckle, intravitreal silicone oil, or perfluorocarbon liquid infusion. All participants provided written informed consent. The study was approved by the Institutional Review Board of the National Taiwan University Hospital and was registered in ClinicalTrials.gov (ID: NCT 04902170). This study adhered to the Declaration of Helsinki.

Patients underwent slit lamp ophthalmoscopy, dilated fundus examination, measurement of refraction (auto kerato-refractometer KR-8800, Topcon Inc), best-corrected visual acuity, AL (Lenstar LS 900, Haag-Streit), intraocular pressure (IOP) (CT-80 Non-Contact Computerized Tonometer; Topcon Corp), and optical coherence tomography (OCT) (RTVue XR Avanti; Optovue Inc). Patients with ERM were indicated for surgery if they experienced metamorphopsia or had visual acuity below 20/50. For those with MTM, surgery was indicated if retinoschisis progressed with visual deterioration or if foveal detachment or an MH were present.

The enrolled eyes were randomized (1:1) by a computer-generated method into the study group (25-gauge vitrectomy probe with a 31.75 mm shaft length [UltraVit 25 Ga 5000 CPM, Alcon]) and the control group (25-gauge vitrectomy probe with a 27 mm shaft length [UltraVit 25+ Ga 7500 CPM, Alcon]) (Fig 1). The sample size calculation was based on the estimated trocar removal rates of 10% in the study group and 40% in the control group, with a statistical power of 85%, a type I error (α) of 0.05, and an anticipated 10% loss to follow-up. Stratified randomization was performed to ensure a balanced distribution of eyes with ALs >28 mm. The patients and the recorders were masked from the results of randomization.

The primary outcomes are the trocar removal rate and the time for core vitrectomy. The secondary outcome is the occurrence of instrument bending. The intraoperative complications were also documented. Inadequate shaft length was defined as the inability to effectively induce posterior vitreous detachment, perform ILM trimming, or remove residual cortical vitreous from the macular surface, necessitating trocar removal to complete the surgery. Additionally, if the position of the sclerotomy wound restricts the vitrectomy probe's ability to efficiently access and treat the lesion, the trocar will be removed to improve probe maneuverability and optimize lesion management. Instrument bending is defined as a visible curvature or deviation from the original straight axis, detectable to the naked eye or under a surgical microscope.

Patients had postoperative follow-ups scheduled for 1 day, 3 days, 1 week, 1 month, 3 months, and 6 months. The presence of vitreous hemorrhage, choroidal detachment, retinal hemorrhage, endophthalmitis, retinal break, RD, transient ocular hypertension (new-onset IOP >21 mmHg), and transient hypotony (new-onset IOP <8 mmHg) were recorded. Additionally, the presence of subconjunctival hemorrhage, subconjunctival air, and wound leakage were recorded at 1 day, 3 days, and 1 week, respectively. Corrected distance visual acuity was measured at 1 month, 3 months, and 6 months. The development of a visually significant cataract requiring surgery was documented. Anatomical success was defined based on the indication of PPV, including closure of MH, removal of ERM, clearing of media opacity, resolution of MTM at foveal and parafoveal regions, and attachment of the retina.

Surgical Technique

One surgeon (T.C.-H.) performed all surgeries under retrobulbar anesthesia using the Alcon Constellation Vision System with a 3-port 25-gauge PPV technique (CONSTELLATION Vitrectomy TOTAL PLUS Pak) and the same settings in both groups. The sclerotomy site was positioned 3.75 mm posterior to the limbus in phakic eyes and 3.25 mm posterior to the limbus in pseudophakic eyes. The aspiration level was up to 500 mmHg with full depression of the foot pedal while maintaining a constant cut rate of 5000 cuts/minutes, and IOP at 25 mmHg.

In patients with significant cataracts, phacoemulsification and posterior chamber intraocular lens implantation were performed concurrently. Posterior vitreous detachment was induced using a vitrectomy probe if necessary, and the posterior hyaloid was removed with or without triamcinolone staining.

For eyes with macular diseases, the ILM was stained with 0.05% indocyanine green. Epiretinal membrane was removed using microforceps, and a bent vitreoretinal blade was used for ILM peeling initiation. In eyes with MTM, a center ILM nonpeeling technique was used.[15] For eyes with MH and macular hole retinal detachment (MHRD), a 270° C-shaped temporal ILM flap measuring 1.5 disc diameters was peeled toward the center of the fovea until 0.5 disc diameters from the center and inverted to cover the MH.[16] Air–fluid exchange was performed using the backflush needle, with or without gas tamponade using 15% perfluoropropane (C_3F_8). Retinal detachment was treated by a 360° peripheral vitrectomy. Internal drainage and air–fluid exchange were performed, followed by gas tamponade with 15% C_3F_8. Finally, the trocars were removed, and the sclerotomy sites were closed by 8-0 Vicryl if there were any signs of leakage.

Statistical Analysis

R (version 4.2.0; R Foundation for Statistical Computing) was used for statistical analyses. Based on an assumed trocar removal rate of 10% in the study group and 40% in the control group, with a 2-tailed alpha of 0.05 and a 90% follow-up rate, each arm required 42 patients to attain a statistical power of 85%. To compare clinical characteristics between the groups, *t* tests or chi-squared tests were used. To investigate the efficacy and safety of the long-shaft vitrectomy probe in eyes with extremely long AL, a subgroup analysis was also performed to compare patients with AL >28 mm and those without. Statistical significance was determined by *P* values <0.05.

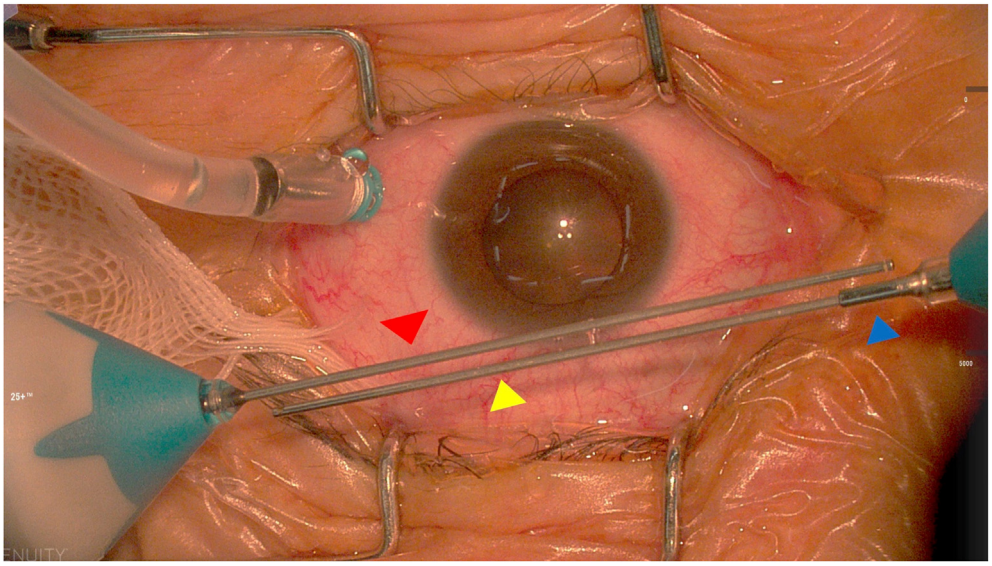


Figure 1. The comparison of the 25 Ga UltraVit and the 25 Ga + UltraVit. The 25 Ga + UltraVit (yellow triangle) has a shorter effective shaft length of 27 mm and is equipped with a stiffness sleeve (indicated by the blue triangle). In contrast, the 25 Ga UltraVit is longer with an effective shaft length of 31.75 mm and does not have a stiffness sleeve.

Results

The study and control group consisted of 44 and 42 eyes, respectively. The mean age was 60.7 ± 9.6 years, and 65 eyes (76%) belonged to female patients. The mean AL was 29.15 ± 2.14 mm. Myopic tractional maculopathy was the most common diagnosis (38%) and was followed by ERM (33%) and MH (27%). These characteristics were comparable between groups (Table 1). Sixteen patients underwent concurrent cataract surgery.

The core vitrectomy time was comparable between groups ($P = 0.172$, t test) (Table 2). In the subgroup analysis of eyes with AL >28 mm, the core vitrectomy time tended to be shorter in the study group (8.1 ± 1.7 vs. 8.8 ± 1.7, $P = 0.114$). More eyes in the control group required trocar removal (odds ratio, 0.02 [0.01−0.11], 67% vs. 5%, $P < 0.001$, chi-square test). In the eyes of the control group with AL >28 mm, 85.7% required trocar removal. There was no significant difference in the trocar removal rate between the groups for individuals with an AL <28 mm ($P = 0.091$). Regarding the secondary outcome, instrument bending was more frequently observed in the study group (odds ratio, 3.42 [1.19−9.90], 36% vs. 14%, $P = 0.036$). The instrument bending occurred mainly in eyes with AL >28 mm, which accounted for 93.8% in the study group and 100% in the control group. The study group showed a significantly higher rate of instrument bending only in those with an AL >28 mm ($P = 0.036$). The total operation time was comparable between groups ($P = 0.867$). In the control group, 24 eyes (57%) were deemed to have inadequate length of the vitrectomy probe to complete surgery, whereas it was noted in only 1 eye (2%) with an AL of 31.92 mm receiving PPV for MTM in the study group ($P < 0.001$). In the control group, 92% of eyes with insufficient shaft length had an AL >28 mm. No case required conversion to 23-gauge instrumentation. As for the safety parameters, 1 eye in the study group experienced hypotony, and another eye in the study group had an iatrogenic retinal break intraoperatively.

Table 1. Clinical Characteristics, Diagnosis, and Preoperative Findings

Parameters	Study	Control
Number of eyes	44	42
Age (yrs)	60.3 ± 10.2	61.2 ± 9.2
Female sex (n, %)	35 (80)	30 (71)
Pseudophakia (n, %)	15 (34)	12 (29)
Axial length (mm)	29.04 ± 2.01	29.26 ± 2.30
Axial length >28 mm	29 (66)	28 (67)
Axial length ≤28 mm	15 (34)	14 (33)
Preoperative intraocular pressure (mmHg)	15.7 ± 3.0	15.1 ± 3.4
Concurrent cataract surgery (n, %)	9 (21)	7 (17)
Diagnosis		
Retinal detachment (n, %)	1 (2)	1 (2)
Epiretinal membrane (n, %)	16 (36)	12 (29)
Full-thickness macular hole (n, %)	4 (9)	7 (17)
Macular hole retinal detachment (n, %)	6 (14)	7 (17)
Myopic tractional maculopathy (n, %)	17 (39)	15 (35)

Table 3 displays the postoperative complications. Neither group exhibited vitreous hemorrhage, choroidal detachment, retinal hemorrhage, endophthalmitis, retinal break, or detachment postoperatively. At postoperative day 1, 33 eyes in the study group and 28 eyes in the control group experienced subconjunctival hemorrhage ($P = 0.854$), and all resolved by postoperative week 1. At postoperative day 1, 2 eyes in the study group and 2 eyes in the control group had transient ocular hypertension ($P = 0.999$), whereas 5 eyes in the study group and 4 eyes in the control group had transient hypotony ($P = 0.999$). The occurrence of ocular hypertension peaked at 1 week after surgery, which could be managed medically. No patients

Table 2. Primary Outcome, Secondary Outcome, and the Intraoperative Complications

Parameters	Study	Control	Difference/Odds Ratio (95% CI)	*P* Value
Primary outcome				
Core vitrectomy time (min)	8.2 ± 1.7	8.7 ± 1.9	−0.5 (−1.3 to 0.2)	0.172
Axial length >28 mm	8.1 ± 1.7	8.8 ± 1.7	−0.7 (−1.6 to 0.2)	0.114
Axial length ≤28 mm	8.3 ± 1.8	8.5 ± 2.2	−0.2 (−1.7 to 1.4)	0.828
Trocar removal (n, %)	2 (5)	28 (67)	0.02 (0.01–0.11)	**<0.001**
Axial length >28 mm	2 (7)	24 (86)	0.01 (0.002–0.07)	**<0.001**
Axial length ≤28 mm	0 (0)	4 (29)	0.08 (0.01–1.55)	0.091
Secondary outcome				
Instrument bending (n, %)	16 (36)	6 (14)	3.43 (1.19–9.90)	**0.036**
Axial length >28 mm	15 (52)	6 (21)	3.93 (1.23–12.53)	**0.036**
Axial length ≤28 mm	1 (7)	0 (0)	3.20 (0.12–85.20)	0.999
Intraoperative findings and complications				
Total operation time (min)	40.7 ± 10.3	39.9 ± 10.1	0.8 (−3.5 to 5.10)	0.717
Wound leakage (n, %)	2 (5)	2 (5)	0.95 (0.13–7.08)	0.999
Choroidal hemorrhage (n, %)	0 (0)	0 (0)	NA	NA
Iatrogenic lens touch (n, %)	0 (0)	0 (0)	NA	NA
Subconjunctival hemorrhage (n, %)	2 (5)	7 (17)	0.24 (0.05–1.22)	0.193
Hypotony (n, %)	1 (2)	0 (0)	2.93 (0.12–73.98)	0.999
Iatrogenic retinal break (n, %)	1 (2)	0 (0)	2.93 (0.12–73.98)	0.999
Length of vitrector (fair: too long: too short)	43 : 0 : 1	18 : 0 : 24	57.3 (7.20–456.5)	**<0.001**
Axial length >28 mm	28 : 0 : 1	6 : 0 : 22	0.01 (0.001–0.9)	**<0.001**
Axial length ≤28 mm	15 : 0 : 0	12 : 0 : 2	6.2 (0.27–141.32)	0.431

CI = confidence interval; NA = not available due to small case number.
Significant *P* values are shown in bold.

with hypotony developed choroidal detachment. All patients completed follow-up at postoperative 3 months, but 2 patients in the study group missed follow-up at postoperative 6 months. At the end of the follow-up period, 7 eyes in the study group and 9 eyes in the control group required cataract surgery ($P = 0.680$). One case with IOP 7 mmHg was noted at postoperative 1 month in the control group.

The anatomical success was comparable between groups (study: 95%, control: 88%, odds ratio, 2.83 [0.52–15.50], $P = 0.394$). In the study group, 2 eyes had unsuccessful outcomes: One patient with MTM achieved fovea flattening but developed rhegmatogenous RD 4 months after the surgery. The patient achieved retinal attachment after intravitreal silicone oil injection and encircling buckle were performed. Another patient with MTM and MHRD had a flattened retina but an unclosed hole after receiving PPV and ILM peeling. The patient received cataract surgery, anterior capsular flap insertion, temporal inverted ILM flap, and intravitreal C_3F_8 injection later, and the MH was closed. In the control group, 5 eyes did not achieve primary anatomical success: 1 patient with rhegmatogenous RD had recurrence 1 month later. Nevertheless, the retina reattached after intravitreal silicone oil injection and encircling buckle were performed. Two patients with ERM who underwent PPV, ERM removal, and ILM peeling developed MH and lamellar MH after surgery, respectively, both of which were closed after receiving an inverted ILM flap. Another patient with MH received PPV and temporal inverted ILM flap but had persistent MH postoperatively, which was closed successfully after inverted ILM was performed again. One patient with MHRD achieved MH closure but had persistent subretinal fluid.

Figure 2 illustrates the visual outcomes after surgery, indicating a substantial improvement in both groups at 3 and 6 months. Only the control group demonstrated significant visual improvement at 1 month postoperatively. However, there was no significant difference in the degree of visual improvement at this time point between the study and control groups (−0.07 ± 0.35 vs. −0.13 ± 0.39, $P = 0.434$, Table 3). The visual outcomes were further analyzed based on different diagnoses. Patients diagnosed with ERM exhibit better preoperative and final visual acuity than those with MTM and MH and associated RD (all $P < 0.001$). However, only those in the study group had significant visual improvement 3 months (−0.15 [−0.27 to −0.01], $P = 0.028$) and 6 months (−0.21 [−0.38 to −0.04], $P = 0.016$) after the surgery, and all of them achieved anatomical success. Eyes with ERM in the control groups did not have significant visual improvement postoperatively (−0.08 [−0.29 to 0.10], $P = 0.340$), possibly related to a lower anatomical success rate 6 months after the surgery (83%). Patients with MTM in both groups showed significant visual improvement at 3 months (study: −0.21 [−0.31 to −0.10] $P < 0.001$; control: −0.27 [−0.53 to −0.01], $P = 0.042$) and 6 months (study: −0.18 [−0.37 to −0.001], $P = 0.049$; control: −0.41 [−0.80 to −0.01], $P = 0.008$). Notably, patients with MH and MHRD did not experience significant visual improvement despite achieving anatomical success.

Discussion

Accessing the posterior pole in highly myopic eyes during PPV is challenging due to the elongated eyeballs and may

Table 3. Outcomes and Postoperative Complications

Time after the Surgery	1 Day			3 Days			1 Wk		
Group	*Study*	*Control*	P *Value*	*Study*	*Control*	P *Value*	*Study*	*Control*	P *Value*
Number of eyes	44	42		44	42		44	42	
Ocular hypertension	2 (5)	2 (5)	0.999	2 (5)	0 (0)	0.495	12 (27)	12 (29)	0.999
Hypotony	5 (11)	4 (10)	0.999	3 (7)	2 (5)	0.999	0 (0)	1 (2)	0.991
Wound leakage	0 (0)	1 (2)	0.991	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Subconjunctival hemorrhage	33 (75)	28 (67)	0.854	34 (77)	29 (69)	0.537	0 (0)	0 (0)	NA
Subconjunctival air	1 (2)	2 (5)	0.967	1 (2)	1 (2)	0.999	0 (0)	0 (0)	NA
Vitreous hemorrhage	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Choroidal detachment	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Retinal hemorrhage	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Endophthalmitis	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Retinal break	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Retinal detachment	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA

Time after Surgery	1 mo			3 mos			6 mos		
Group	*Study*	*Control*	P *Value*	*Study*	*Control*	P *Value*	*Study*	*Control*	P *Value*
Number of eyes	44	42		44	42		42	42	
Visual improvement	−0.07 ± 0.35	−0.13 ± 0.39	0.434	−0.17 ± 0.31	−0.18 ± 0.51	0.884	−0.22 ± 0.37	−0.24 ± 0.45	0.748
Complications (n, %)									
Ocular hypertension	7 (16)	4 (10)	0.573	5 (11)	6 (14)	0.934	5 (12)	3 (7)	0.710
Hypotony	0 (0)	1 (2)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Vitreous hemorrhage	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Choroidal detachment	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Retinal hemorrhage	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Endophthalmitis	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	NA
Retinal break	0 (0)	1 (2)	NA	0 (0)	0 (0)	NA	1 (2)	0 (0)	NA
Retinal detachment	0 (0)	1 (2)	NA	0 (0)	0 (0)	NA	1 (2)	0 (0)	NA

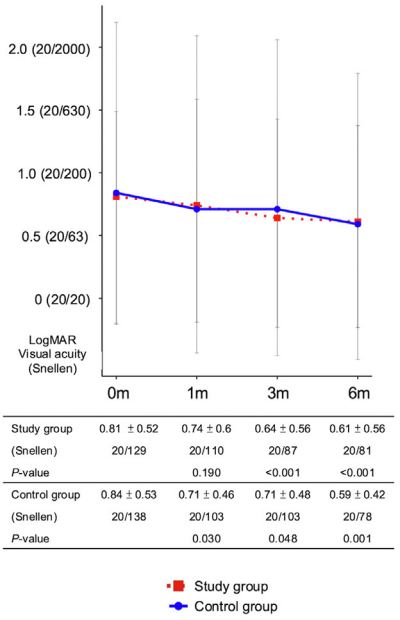
NA = not available due to small case number.

require specialized surgical instruments.[17] The adequate shaft length of a vitrectomy probe is crucial for completing critical steps, such as inducing posterior vitreous detachment and trimming the ILM flap, in highly myopic eyes. Our study assessed the effectiveness and safety of an alternative 25-gauge vitrectomy probe with a longer shaft length of 31.75 mm (UltraVit 25 Ga) compared with the standard 25-gauge vitrectomy probe with a shaft length of 27 mm (Ultravit 25 Ga+) in highly myopic eyes. There were no significant differences between groups in terms of the duration of surgery and incidence of complications. Patients with ERM had better visual outcomes using the long-shaft vitrectomy probe. Using long-shaft vitrectomy probes reduced the need for trocar removal at the cost of more instrument bending. The differences between groups were more notable in eyes with AL >28 mm.

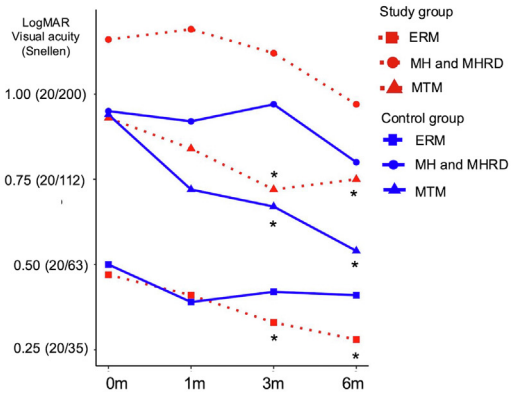
The literature highlights an issue in PPV where standard instruments are too short to reach the posterior pole in highly myopic eyes.[9,10,17,18] Various methods have been proposed, including using 20- or 23-gauge instrumentation with longer effective shaft lengths, removing the trocar to increase effective shaft length, or reducing IOP and indenting the scleral wound to allow instruments to reach the posterior pole.[9] However, these methods carry potential problems. Scleral indentation may cause wound leakage, hypotony, and corneal folding, while verticalization of the instrument can result in the surgeon's finger touching the viewing system, obscuring the view, and making surgical maneuvers more challenging.[13]

A recent proposal by Iwama et al suggests extending the limbus-to-cannula distance to 6 mm in eyes with AL >31 mm,[17] based on the observation that highly myopic eyes with AL >27 mm have the posterior edge of pars plana >6 mm away from the limbus.[17,19] This adjustment can reduce the distance between the cannula and fovea by 1.22 mm.[17] Hirono et al used anterior segment OCT to determine the optimal site for sclerotomy, with a mean entry site length of 6.2 mm from the limbus in a group of patients with a mean AL of 29.2 mm.[20] Al Sabti and Raizada proposed posterior scleral indentation to shorten the operating distance in eyes with AL >34 mm.[21] Teixeira et al used the cosine law to estimate the appropriate entry site and instrument shaft length based on the distance between the sclerotomy site and fovea.[18] These approaches highlight the importance of individualized treatment based on AL to optimize surgical outcomes and minimize complications.

Another possible solution is instruments with longer shaft lengths. Gao et al introduced long-shaft forceps with different shaft lengths for performing ILM peeling in highly myopic eyes.[14] The 37-mm and 39-mm long-shaft forceps are appropriate for eyes with AL around 28 to 30 mm and longer, respectively.[14] Lytvynchuk et al used modified curved aspiration cannulas and forceps during 25-gauge PPV in



Study group	0.81 ± 0.52	0.74 ± 0.6	0.64 ± 0.56	0.61 ± 0.56
(Snellen)	20/129	20/110	20/87	20/81
P-value		0.190	<0.001	<0.001
Control group	0.84 ± 0.53	0.71 ± 0.46	0.71 ± 0.48	0.59 ± 0.42
(Snellen)	20/138	20/103	20/103	20/78
P-value		0.030	0.048	0.001



Study group				
ERM	0.47 ± 0.26	0.41 ± 0.37	0.33 ± 0.21	0.28 ± 0.28
(Snellen)	20/59	20/51	20/43	20/38
MH & MHRD	1.16 ± 0.59	1.19 ± 0.69	1.12 ± 0.73	0.97 ± 0.70
(Snellen)	20/289	20/310	20/264	20/187
MTM	0.93 ± 0.52	0.84 ± 0.59	0.72 ± 0.52	0.75 ± 0.54
(Snellen)	20/170	20/138	20/105	20/112
Control group				
ERM	0.50 ± 0.19	0.39 ± 0.33	0.42 ± 0.32	0.41 ± 0.38
(Snellen)	20/63	20/49	20/53	20/51
MH & MHRD	0.95 ± 0.49	0.92 ± 0.40	0.97 ± 0.46	0.80 ± 0.40
(Snellen)	20/178	20/166	20/187	20/126
MTM	0.94 ± 0.61	0.72 ± 0.48	0.67 ± 0.51	0.54 ± 0.43
(Snellen)	20/174	20/105	20/94	20/69

Figure 2. The visual and anatomical outcomes of the patients with different diagnoses. Left, At postoperative 3 months, patients in the study group experienced a statistically significant improvement in logMAR visual acuity from 0.81 to 0.64 (Snellen: 20/129 to 20/87), which further improved to 0.61 (Snellen: 20/81) at postoperative 6 months. Patients in the control group showed significant improvement from 0.84 to 0.71 (Snellen: 20/138 to 20/103) at postoperative 1 and 3 months and continued to improve to 0.59 (Snellen: 20/78) at postoperative 6 months. Right, Eyes with ERM exhibited the best baseline visual acuity, followed by those with MTM, while eyes with MH and MHRD exhibited the worst baseline visual acuity. After surgical intervention, eyes with MH and MHRD did not exhibit a statistically significant visual improvement. Conversely, eyes with ERM or MTM showed a significant improvement in visual acuity from postoperative 3 months in the study group, except those with ERM in the control group. A *P* value <0.05 was considered statistically significant and denoted by a pound sign on the figure. The postoperative visual acuity was compared with the preoperative visual acuity and was shown in bold if there was a significant improvement. ERM = epiretinal membrane; logMAR = logarithm of the minimum angle of resolution; MH = macular hole; MHRD = macular hole retinal detachment; MTM = myopic tractional maculopathy.

highly myopic eyes to prevent any contact between the surgeon's fingers or instruments with the viewing system while verticalizing the instrument.[13] A long-shaft vitrectomy probe is crucial for successful vitrectomy. An effective technique for inducing posterior vitreous detachment involves the use of vitrectomy probes for active aspiration at the epipapillary region.[22] Additionally, vitrectomy probes can be utilized for trimming the ILM flap tailored for covering MH.[23–25] Both the 20- and 23-gauge vitrectomy probes had a 31.75 mm shaft length. However, as the diameter of the vitrectomy probe decreased in the small-gauge system, the instrument's stiffness decreased. To address this issue, either a metal sleeve was added to increase stiffness or the shaft length was shortened.[26] The UltraVit 25+ Ga has such stiffness sleeves with an effective shaft length of 27.00 mm. The UltraVit 27+ Ga has an even shorter effective shaft length of 26 mm, which may make it difficult to reach the posterior pole in highly myopic eyes.

The UltraVit 25 Ga probe, due to the absence of a stiffness sleeve, offers a longer effective probe length compared with the UltraVit 25 Ga+, making it comparable to the UltraVit 20 Ga and 23 Ga probes. This study aimed to evaluate the safety and efficacy of the UltraVit 25 Ga in comparison to the UltraVit 25 Ga+. The core vitrectomy time was chosen as the primary outcome to assess efficiency, revealing comparable results between the 2 groups. However, the UltraVit 25 Ga demonstrated a distinct advantage in eyes with an AL >28 mm, where it tended to achieve a shorter core vitrectomy time. Additionally, the trocar removal rate was designated as a primary outcome, as it serves as a surrogate indicator for evaluating the adequacy of vitrectomy shaft length in highly myopic eyes. The control group had a higher frequency of trocar removal and inadequate effective shaft length in eyes with AL >28 mm. In contrast, the UltraVit 25 Ga allowed for more delicate trimming of the ILM flap and meticulous vitreous removal due to its extra effective shaft length. Therefore, for eyes with AL >28 mm, using the UltraVit 25 Ga may be more appropriate to achieve optimal surgical outcomes. Despite its advantages, the UltraVit 25 Ga had a more flexible shaft, making it more prone to bending. Hence, future designs of vitrectomy probes should address this issue to ensure optimal surgical outcomes. Our analysis of safety outcomes showed that both groups had low complications, such as endophthalmitis, vitreous hemorrhage, and choroidal detachment, up to 6 months after surgery. No cases of iatrogenic lens

touch were reported in either group. In the study group, 1 patient experienced an iatrogenic small retinal break at the inferior periphery, which was treated with endolaser. During postoperative follow-up, the patients did not develop RD and achieved a final OCT corrected visual acuity of 20/25. Another patient in the study group experienced hypotony intraoperatively; however, the IOP on postoperative day 1 was 11.5 mmHg, and no choroidal detachment was observed. The proportion of patients requiring subsequent cataract surgery did not differ between the groups at the end of the follow-up. Therefore, the safety profile of the UltraVit 25 Ga is comparable to that of the UltraVit 25 Ga+.

In this study, the primary anatomical success rate was 92%, and significant visual improvement was achieved. The outcome could vary with the presenting diagnosis. Highly myopic eyes with MH, especially those with an AL >30 mm and complicated with MTM, tend to have less favorable outcomes.[6,27–30] The primary MH closure rate was 92% in our cohort; however, the visual improvement was not significant due to the small cohort. The eyes with MH also take more time for visual rehabilitation and could have visual improvement up to 1 year after the surgery.[16,31] Conversely, eyes with ERM or MTM have better outcomes than those with MH. During postoperative follow-up, substantial visual improvement was observed at 3 and 6 months, except for eyes with ERM in the control group. The suboptimal visual outcome in controls with ERM may be explained by a lower anatomical success rate. Two cases developed MH and lamellar MH, respectively, and received secondary PPV within 6 months of the initial PPV. Although not achieving statistical significance, the study group had better anatomical outcomes than the control group (95% vs. 88%). The longer vitrectomy probe provided better access to the retinal surface to remove residual cortical vitreous and to remove indocyanine green and elevate the ERM margins, which may benefit the anatomical outcomes.

This study has several limitations. First of all, the outcome measurements in this study involved subjectivity. The decision to remove the trocar varied among surgeons, influenced by their individual preferences and experiences. Additionally, the assessment of shaft length adequacy and trocar bending was inherently subjective, relying on the surgeon's expertise, intraoperative techniques, and case complexity. Since the study was conducted by a single surgeon, future investigations with larger cohorts involving multiple surgeons could provide additional validation of these subjective findings. The complete masking of the surgeon from randomization outcomes was challenging, as it could be inferred from the varying vitrectomy probe designs. Despite efforts to minimize potential bias by masking patients and outcome assessors, the surgeon's awareness of the instrument used may still introduce bias in outcome evaluation. To enhance objectivity in future research, the incorporation of advanced assessment tools, such as intraoperative OCT, could provide a quantitative evaluation of vitreous removal completeness and reduce subjective interpretation. Finally, the patients underwent follow-ups for 6 months. Certain patients may require additional time or subsequent cataract surgery to attain complete visual rehabilitation. Extending the follow-up period is worth considering in future studies. Furthermore, the current study included a diverse group of patients with various diagnoses. Subsequent research involving patients sharing a common diagnosis may provide additional insights into the advantages of utilizing the long-shaft vitrectomy probe.

In conclusion, this study demonstrates the safety and effectiveness of 25Ga UltraVit, a vitrectomy probe with a longer shaft length, in highly myopic eyes undergoing PPV. Especially in eyes with an AL >28 mm, the use of a long-shaft vitrectomy probe can provide better access to the posterior pole for complete ILM trimming and posterior hyaloid removal, potentially avoiding the need for trocar removal and improving visual outcome. It is important to note that the sleeveless design of the instrument led to a higher instrument bending rate, which should be taken into consideration in future designs for instruments used in highly myopic eyes.

Data Availability

Data are available from the corresponding author upon reasonable request.

Acknowledgments

The authors would like to express their thanks to the staff of the National Taiwan University Hospital-Statistical Consulting Unit (NTUH-SCU) for statistical consultation and analyses.

Footnotes and Disclosures

Originally received: December 12, 2024.
Final revision: February 4, 2025.
Accepted: April 30, 2025.
Available online: June 16, 2025. Manuscript no. XOPS-D-24-00552.

[1] Department of Ophthalmology, National Taiwan University Hospital, College of Medicine, National Taiwan University, Taipei City, Taiwan.

[2] Department of Ophthalmology, National Taiwan University Hospital Hsin-Chu Branch, Hsinchu, Taiwan.

[3] Department of Ophthalmology, An Nan Hospital, China Medical University, Tainan, Taiwan.

[4] Department of Ophthalmology, Shuang Ho Hospital, Taipei Medical University, New Taipei City, Taiwan.

[5] Department of Ophthalmology, Fu Jen Catholic University Hospital, New Taipei City, Taiwan.

[6] Universal Eye Clinic, Taipei, Taiwan.

[7] Department of Ophthalmology, Cardinal Tien Hospital, New Taipei City, Taiwan.

The manuscript was presented as an ePoster at the American Academy of Ophthalmology Annual Meeting, November 3-6, San Francisco, California, 2023.

Disclosure(s):

The Article Publishing Charge (APC) for this article was paid by National Taiwan University Hospital, College of Medicine, National Taiwan University.

All authors have completed and submitted the ICMJE disclosures form.

The authors have no proprietary or commercial interest in any materials discussed in this article.

This study is supported by Alcon Medical Affairs, IIT#57166923 grant, and the Ministry of Health and Welfare, Taiwan (MOHW110-TDU-B-211-124002).

HUMAN SUBJECTS: Human subjects were included in this study. All participants provided written informed consent. The study was approved by the Institutional Review Board of the National Taiwan University Hospital and was registered in ClinicalTrials.gov (ID: NCT 04902170). This study adhered to the Declaration of Helsinki.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Hsia, Wang, Chien-Jung Huang, Ma, Hung, Chen, Ho

Data collection: Hsia, Lee, Tsui, Yang, Bo-Da Huang, Ko, Ho

Analysis and interpretation: Hsia

Obtained funding: Ho

Overall responsibility: Hsia, Lee, Tsui, Wang, Chien-Jung Huang, Ma, Hung, Chen, Yang, Bo-Da Huang, Ko, Ho

Abbreviations and Acronyms:
AL = axial length; **ERM** = epiretinal membrane; **ILM** = internal limiting membrane; **IOP** = intraocular pressure; **MH** = macular hole; **MHRD** = macular hole retinal detachment; **MTM** = myopic tractional maculopathy; **PPV** = pars plana vitrectomy; **RD** = retinal detachment.

Keywords:
High myopia, Long-shaft vitrectomy probe, Myopic tractional maculopathy, Epiretinal membrane, Pars plana vitrectomy.

Correspondence:
Tzyy-Chang Ho, MD, Department of Ophthalmology, National Taiwan University Hospital, College of Medicine, National Taiwan University, No. 7, Chung-Shan S. Rd., Taipei 10002, Taiwan. E-mail: hotchang@ntu.edu.tw.

References

1. Bourne RR, Stevens GA, White RA, et al. Causes of vision loss worldwide, 1990-2010: a systematic analysis. *Lancet Glob Health*. 2013;1:e339–e349.
2. Holden BA, Fricke TR, Wilson DA, et al. Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology*. 2016;123:1036–1042.
3. Ohno-Matsui K, Lai TY, Lai CC, Cheung CM. Updates of pathologic myopia. *Prog Retin Eye Res*. 2016;52:156–187.
4. Fricke TR, Jong M, Naidoo KS, et al. Global prevalence of visual impairment associated with myopic macular degeneration and temporal trends from 2000 through 2050: systematic review, meta-analysis and modelling. *Br J Ophthalmol*. 2018;102:855–862.
5. Ikuno Y. Overview of the complications of high myopia. *Retina*. 2017;37:2347–2351.
6. Alkabes M, Pichi F, Nucci P, et al. Anatomical and visual outcomes in high myopic macular hole (HM-MH) without retinal detachment: a review. *Graefes Arch Clin Exp Ophthalmol*. 2014;252:191–199.
7. Cheong KX, Xu L, Ohno-Matsui K, et al. An evidence-based review of the epidemiology of myopic traction maculopathy. *Surv Ophthalmol*. 2022;67:1603–1630.
8. Fang D, Liang J, Chen S, et al. Surgical outcomes of myopic foveoschisis according to the ATN classification system. *Ophthalmol Ther*. 2023;12:71–85.
9. Coppola M, Rabiolo A, Cicinelli MV, et al. Vitrectomy in high myopia: a narrative review. *Int J Retina Vitreous*. 2017;3:37.
10. Singh A, Fawzi AA, Stewart JM. Limitation of 25-gauge vitrectomy instrumentation in highly myopic eyes. *Ophthalmic Surg Lasers Imaging*. 2007;38:437–438.
11. Woo SJ, Park KH, Hwang JM, et al. Risk factors associated with sclerotomy leakage and postoperative hypotony after 23-gauge transconjunctival sutureless vitrectomy. *Retina*. 2009;29:456–463.
12. McBrien NA, Cornell LM, Gentle A. Structural and ultrastructural changes to the sclera in a mammalian model of high myopia. *Invest Ophthalmol Vis Sci*. 2001;42:2179–2187.
13. Lytvynchuk LM, Sergiienko A, Richard G. Modified curved aspiration cannulas and end-gripping forceps for 25-GAUGE vitrectomy on highly myopic eyes. *Retina*. 2015;35:2660–2663.
14. Gao X, Ikuno Y, Nishida K. Long-shaft forceps for membrane peeling in highly myopic eyes. *Retina*. 2013;33:1475–1476.
15. Ho TC, Yang CM, Huang JS, et al. Long-term outcome of foveolar internal limiting membrane nonpeeling for myopic traction maculopathy. *Retina*. 2014;34:1833–1840.
16. Ho TC, Ho A, Chen MS. Vitrectomy with a modified temporal inverted limiting membrane flap to reconstruct the foveolar architecture for macular hole retinal detachment in highly myopic eyes. *Acta Ophthalmol*. 2018;96:e46–e53.
17. Iwama Y, Ikeda T, Nakashima H, et al. Extending the limbus-to-cannula distance to 6.0 mm during pars plana vitrectomy in highly myopic eyes. *Retina*. 2022;42:1199–1202.
18. Teixeira A, Salaroli C, Fuganti RM, Casella AMB. Use of the cosine law to choose the correct forcepts shaft length for macular surgery in highly myopic eyes. *Retina*. 2023;43:2166–2169.
19. Sudhalkar A, Chauhan P, Sudhalkar A, Trivedi RH. Pars plana width and sclerotomy sites. *Ophthalmology*. 2012;119:198–199.e1-3.
20. Hirono K, Inoue M, Tanaka S, et al. Anterior segment OCT in determination of entry site for vitrectomy in highly myopic eyes. *Retina*. 2023;43:733–738.
21. Al Sabti K, Raizada S. Posterior Scleral Indentation in long axial length eyes (axial length more than 34 mm) to assist in pars plana vitrectomy and internal limiting membrane peeling in full thickness macular holes. *Retina*. 2023;43:705–710.
22. Ellabban AA, Barry R, Sallam AAB. Surgical induction of posterior vitreous detachment using combined sharp dissection and active aspiration. *Acta Ophthalmol*. 2016;94:e524–e525.
23. da Silva Tavares Neto JE, Coelho IN, Jorge R, et al. Pedicle internal limiting membrane flap technique for very large macular holes: a preliminary report. *Int J Retina Vitreous*. 2020;6:43.
24. Bleidißel N, Friedrich J, Klaas J, et al. Inverted internal limiting membrane flap technique in eyes with large idiopathic full-thickness macular hole: long-term functional and morphological outcomes. *Graefes Arch Clin Exp Ophthalmol*. 2021;259:1759–1771.

25. Hung J-H, Horng Y-H, Chu H-C, et al. Tailored internal limiting membrane flap technique for primary macular hole. *Graefes Arch Clin Exp Ophthalmol*. 2020;258:759−766.
26. de Oliveira PRC, Berger AR, Chow DR. Vitreoretinal instruments: vitrectomy cutters, endoillumination and wide-angle viewing systems. *Int J Retina Vitreous*. 2016;2:28.
27. Chen SN, Yang CM. Inverted internal limiting membrane insertion for macular hole-associated retinal detachment in high myopia. *Am J Ophthalmol*. 2016;162:99−106.e1.
28. Chen SN, Hsieh YT, Yang CM. Multiple free internal limiting membrane flap insertion in the treatment of macular hole-associated retinal detachment in high myopia. *Ophthalmologica*. 2018;240:143−149.
29. Yuan J, Zhang LL, Lu YJ, et al. Vitrectomy with internal limiting membrane peeling versus inverted internal limiting membrane flap technique for macular hole-induced retinal detachment: a systematic review of literature and meta-analysis. *BMC Ophthalmol*. 2017;17:219.
30. Hsia Y, Lee CY, Ho TC, et al. The development and evolution of lamellar macular hole in highly myopic eyes. *Eye (Lond)*. 2023;37:1170−1177.
31. Steel DH, Lotery AJ. Idiopathic vitreomacular traction and macular hole: a comprehensive review of pathophysiology, diagnosis, and treatment. *Eye (Lond)*. 2013;27 Suppl 1(Suppl 1):S1−S21.